

Smash Hit Scoreboard

An autonomous scoreboard that immerses you in the game

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Chapter 2 - Project Description

2.1 Motivation and Background

Racquet sports are fun and competitive games that benchmark reaction time and targeting skills. These games play out in a very fast-paced and intricate manner, with small adjustments leading to completely different outcomes. These games bring out the most intense side of each player, pushing them to be the victors.

However, one key issue many players face while playing is keeping track of the score. It is something that may require a third person to act as a referee because of how easy it is to forget while playing. Players can also run into issues of getting their scores confused or not properly tallying points after rallies.

This motivated us to try to find a way to delegate the responsibility of tracking the score of a Ping Pong match to a device that was able to understand the current state of the game and update the scoreboard accordingly.

While tracking the gameplay for such sports can prove to be a challenging task, as shown by controversial calls at Wimbledon back in 2025 [1], we believe the benefits can be massive, especially in competitive spheres. Nevertheless, our intention is not to design this system only for the most advanced players, but instead to cover a wide range of skill levels, ranging from beginners to hobbyists who want to spend less time counting points and more time playing.

Thus, the challenge for us as a team is to come up with a solution that is not only useful to every player regardless of ability, but also simple to use and accurate enough that it is worth the time to set up.

To ensure we achieve our project goals, our progress will be evaluated by a faculty review committee. Dr. Yogesh Singh Rawat and Dr. Shruti Vyas will supervise the software development, Dr. Nazanin Rahnavard and Dr. Saleem Sahawneh will oversee the hardware design, and Dr. Maryam Mahmoodi will review the optical systems.

2.2 Existing Products and Technologies

The following are some relevant projects, tools, and products that made us confident in the feasibility of our idea:

2.2.1 Table Tennis Tracker by Taylor Pritchard, Pol Rosello, and Frank Xie

This system was developed as a Cornell University "Final Project" [2]. This project employed a single camera to track the entire state of the game and was able to achieve over 90% accuracy. While these results were promising, they were attained under specific circumstances that might not be representative of its normal behavior. This project confirms that having a single camera tracking the game is viable and motivates us to be able to get an accurate score for our scoreboard using a single camera.

2.2.2 TableTennisTracker by Rahul Jain

Throughout our preliminary research, we found a project that takes a similar approach to our initial concept for tracking the state of the game [3]. Also inspired by Table Tennis Tracker (Section 2.2.1), this project employed a simple phone camera back in 2022 to develop a similar solution. Being able to do this with a simple phone camera supports our low-cost hardware and helped our decision to target a consumer-grade camera instead of a dedicated high-speed camera.

2.2.3 Ultralytics YOLO models

The YOLO family of computer vision models by Ultralytics provides a wide range of options for object tracking. In particular, they recommend employing "a trained Detect, Segment, Pose, or OBB model such as YOLO26n, YOLO26n-seg, YOLO26n-pose, or YOLO26n-obt." [4] .

Further research also made us aware of the possibility of running one of their models on a Raspberry Pi [5], establishing this as a potential path for executing computer vision algorithms on affordable hardware.

2.3 Goals

The following sections cover key high-level outcomes the project requires in order to be a successful product. These are split into three categories. Basic and advanced goals must be finished in order to satisfy the project's requirements, while stretch goals depend on timelines and budget constraints.

2.3.1 Basic Goals

- Design a digital scoreboard to keep track of the score with manual controls. These include increasing or decreasing a player's score and resetting the tally.
- Develop a laser grid, parallel to the plane of the table, to determine when and where the ping pong ball hits the table.
- Inform players via a simple audio or visual cue when a point is scored.

2.3.2 Advanced Goals

- Automatically update the scoreboard using software with >70% accuracy based on gameplay without the need for user input in real time.
- Allow customization of the game rules, such as setting the number of points to win.

2.3.3 Stretch Goals

- Option to record the game and recreate it in a 3D environment for analysis.
- Provide post-game analysis for each player. These include top ball speed, number of hits, longest streak, and more.
- Provide audio and/or visual feedback tied to the game's state, so players understand without having to pause the game (full score, game over, who scored).

2.4 Objectives

The following sections cover key low-level targets the team is looking to achieve to satisfy its project's goals. These are split into three categories. Basic and advanced objectives must

be finished to satisfy the project's requirements, while stretch objectives depend on timelines and budget constraints.

2.4.1 Basic Objectives

- Implement a computer-vision-assisted system that analyzes >30 frames per second of the game to update the scoreboard. The algorithm should be able to understand the rules of the game and update the tally accordingly.
- The system should be able to tell which user hit the ping pong ball last to correctly update the scoreboard.
- Implement a game state machine that uses ball position/direction changes to detect bounces and hits.
- Select a camera with sufficient resolution and frame rate to see the full table with minimal blind spots.
- Implement connections between all the systems (laser grid, camera, MCU, MPU, power, illumination, buzzers/LEDs, scoreboard).
- Track the position of each bounce using solely a laser grid within 3-5 cm of accuracy.
- Be able to power all the connected systems without an external power supply

2.4.2 Advanced Objectives

- Add audio/visual cues triggered by specific state-machine transitions.
- Reduce motion blur/tracking loss on fast shots by tuning frame rate and exposure.
- Create a laser array that determines if one player hits the net, and award points according to where the ball lands after hitting the net.
- Implement a time multiplex on the laser grid to lessen the load on the MCU
- Maintain accurate ball detection with sufficient lighting conditions and background clutter.

2.4.3 Stretch Objectives

- An instant replay of the last rally trajectory.
- Be able to control the system wirelessly as well as with physical switches. Wirelessly, an option would be through an IR remote or a phone app.
- Log per-rally statistics (speed, hit count, streak length) and build a simple post-game summary.
- Extend state machines and tracking logic to support a second game.
- Stand-alone device that does not need a computer plugged in to operate.

2.5 Features, Functionalities and The Customer

The following is a list of both features and functions the project requires to succeed. The list covers basic, advanced, and stretch goals and objectives, so not all may be present in the final prototype to be demoed in Senior Design II. Furthermore, this section also covers our understanding of who the primary user of our project will be.

2.5.1 Features

2.5.1.1 Basic Features

- Automatic Score tracking for a full match.
- Real-time ball and player detection from one or multiple cameras.
- Live scoreboard display.
- Manually able to correct the score.
- Laser grid for detection of the ball hitting the table.

2.5.1.2 Advanced Features

- Audio/visual cues when a point is scored or when the match is over.
- Self-contained power system for all subsystems.
- Game audio feedback (Serve, Point, Game over).
- Net hit detection with correct points awarded.

2.5.1.3 Stretch Features

- Customizable rules
- Instant Replay of last rally trajectory
- Live game data
- 3D mapping of rally
- Post-game per-player stats (hit count, longest streak).

2.5.2 Functions

2.5.2.1 Basic Functions

- Detect and track ball position and direction
- Detect bounces and paddle hits from ball trajectory changes
- Real laser grid to register contact between the ball and table to get positioning
- Run game state machine (serve, rally, point, game, match)
- Apply scoring rules and update tally
- Drive scoreboard display and read manual inputs
- Trigger buzzer/LED on points scored
- Interface MPU with camera, laser grid, scoreboard, and LEDs/buzzers
- Regulate and distribute power to all subsystems

2.5.2.2 Advanced Functions

- Detect net contact with laser array and correctly give point
- Time-multiplex laser grid emitters/receivers

2.5.2.3 Stretch Functions

- Store and render last rally trajectory
- Map state machine transitions to specific audio/visual outputs
- Receive commands over IR/wireless link
- Load user-created parameters into state machine
- Aim laser pointer to last bounce coordinates
- Run full pipeline on embedded hardware without PC
- Compute and store per-rally data

2.5.3 The Customer

Understanding the customer is one of the most complex tasks of any project. There are multiple factors that go into what a target audience would be: budget, accuracy, and ease of use. Accounting for these factors, we have defined two kinds of “ideal” customers who we think would be highly interested in our solution.

The primary source of customers for this product would be recreational players or “hobbyists”. These are people who play in their homes, garages, dorms, offices, and communal spaces and are already invested in the sport. They can play the game without arguing over points. Not only that, but it will also give users access to more information about their play session, making the experience much more immersive.

The second type of customer would be table tennis clubs and recreation centers that run casual play and/or small, friendly tournaments. Our solution could be employed to facilitate such events while also helping them stand out with visual and audio feedback.

2.6 Required Specifications

A key aspect of the Senior Design project is to define clear, achievable, and measurable specifications. The following two tables determine our objectives on both the software and component side of development. The team must achieve the specifications highlighted in yellow and demo them during Senior Design II in order to successfully pass the class.

2.6.1 Overall System Specification

The following is a list of the overall system specifications the team aims to achieve.

Table 2.6.1: System Specifications

#	Parameter	Description	Specification
1	Bounce Position Tracking Accuracy with camera system	How precisely does the camera system identify where the ball contacts the table’s surface.	~5-10 cm with camera
2	Bounce Position Tracking Accuracy with laser system	How precisely does the laser system identify where the ball contacts the table’s surface.	~3-5 cm with laser grid
3	Bounce Detection Latency	Time from when the ball makes contact with the table to the events being registered by the system.	< 1 second
4	Scoring Accuracy	Percentage of points correctly identified and tallied without manual correction.	≥ 70% of points correctly identified
5	Score Update Latency	Time from a point being scored to the scoreboard	< 5 seconds

		display updating to reflect the new tally.	
6	Power Distribution	Total system power draw and voltage stability across all subsystems.	< 30 Watts total
7	Portability	Total weight and footprint of the assembled system for setup and transport.	< 20 pounds

2.6.2 Component Specifications

The following is a list of the relevant components and their target specifications.

Table 2.6.2: Component Specifications

#	Component	Parameter	Description	Specification
1	Camera	Frames per second (FPS)	Frame rate needed to capture ball movement without excessive motion blur.	> 30 FPS
2	MPU	Power consumption (W)	Voltage level and power draw of the MPU driving sensors and outputs.	4W-15W
3	Lasers	Multiplexing rate	Laser wavelength for detection and the switching rate of the time-multiplexed grid.	≥ 1 kHz scan rate
4	Display	Update time (ms)	Time for the scoreboard to visually refresh after receiving an updated score.	< 1000 ms
5	Buzzer	Volume (dB)	Audible output level and pitch loud enough to notice over gameplay noise.	70-85 dB @ 1 meter
6	Power Supply	Max current (A)	Maximum current the supply must deliver across all subsystems under peak load.	< 6A @ 5V
7	Photodetectors	Response time (S)	Time for a photodetector to register a change in laser beam state.	< 1 ms

8	LED Spotlight (Replay)	Accuracy of where the ping pong ball hits on the table	Precision with which the system illuminates the bounce locations for feedback.	Matches bounce tracking accuracy (~5-10 cm)
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2.7 System Diagrams

A system diagram looks to decompose a complex idea such as the one behind this project into the smallest possible workable chunks. The following block diagrams illustrate this on both the hardware and software side.

2.7.1 Hardware Block Diagram(s)

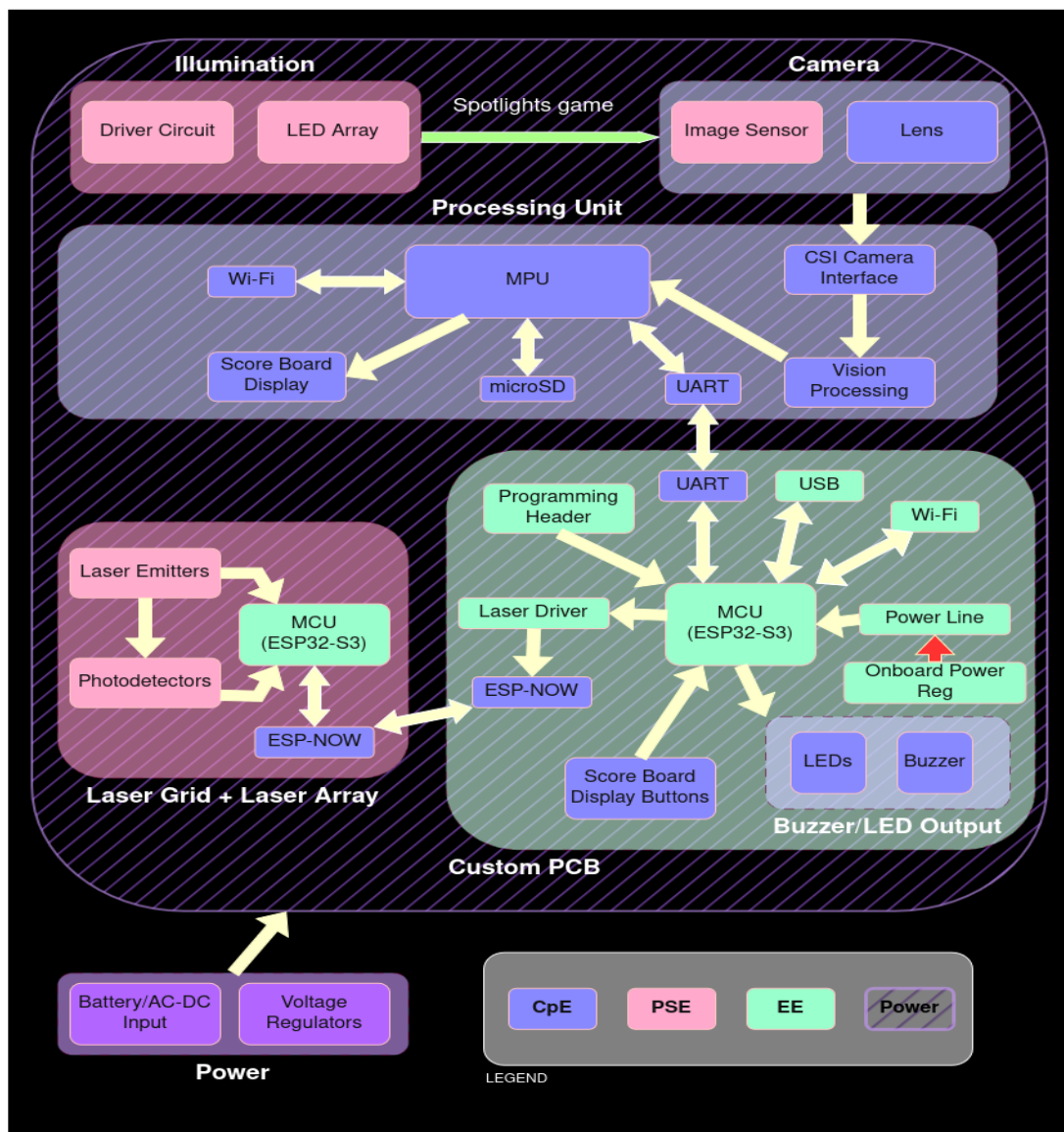


Figure 2.7.1: Hardware Block Diagram

2.7.2 Software Diagram/Flowchart

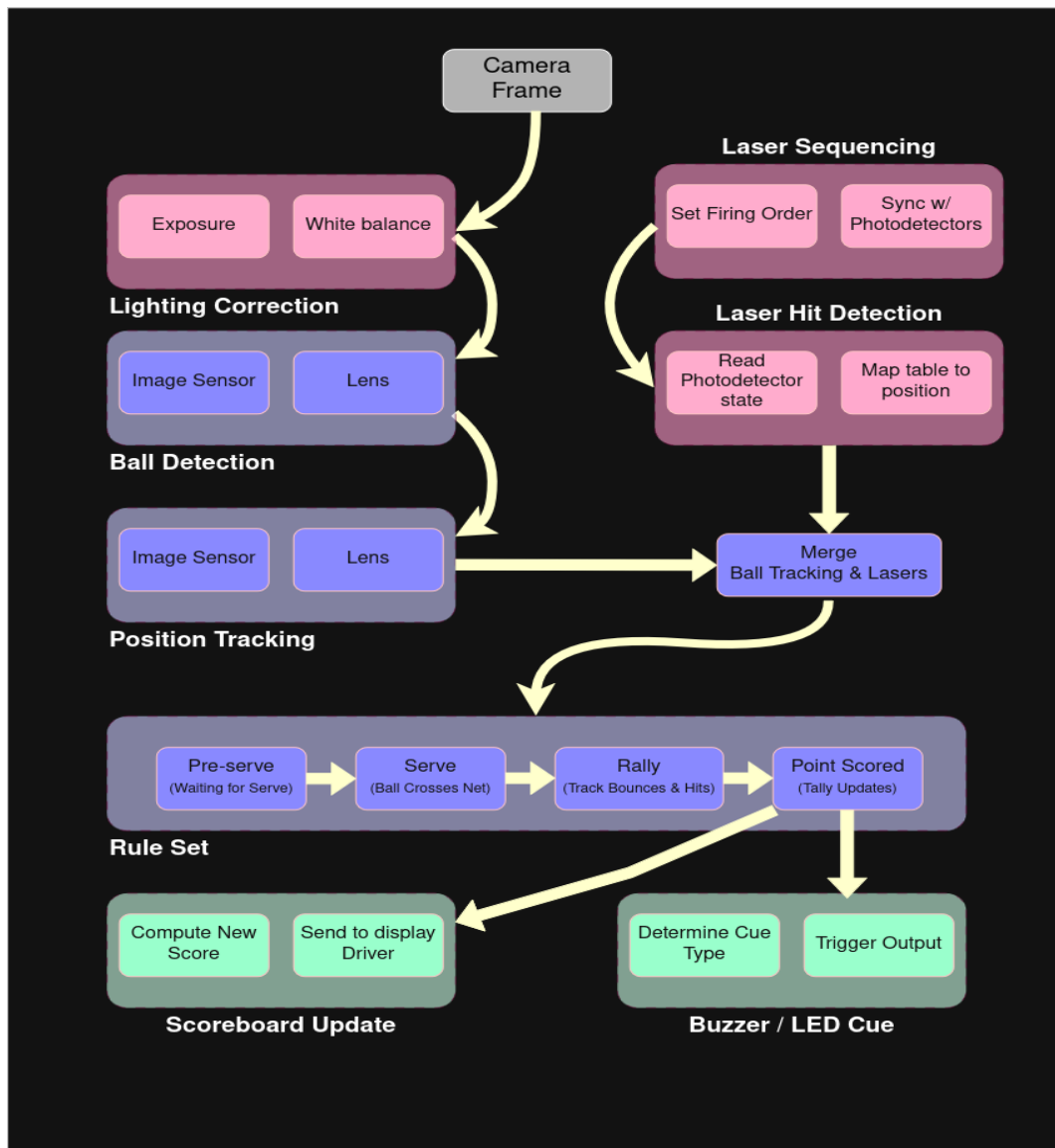


Figure 2.7.2: Software Block Diagram

2.7.3 Prototype Illustration / Blueprint

The following illustrations are meant to provide a wide range of possibilities regarding the final design for our solution. The final design will depend on budget, repairability, and other factors.

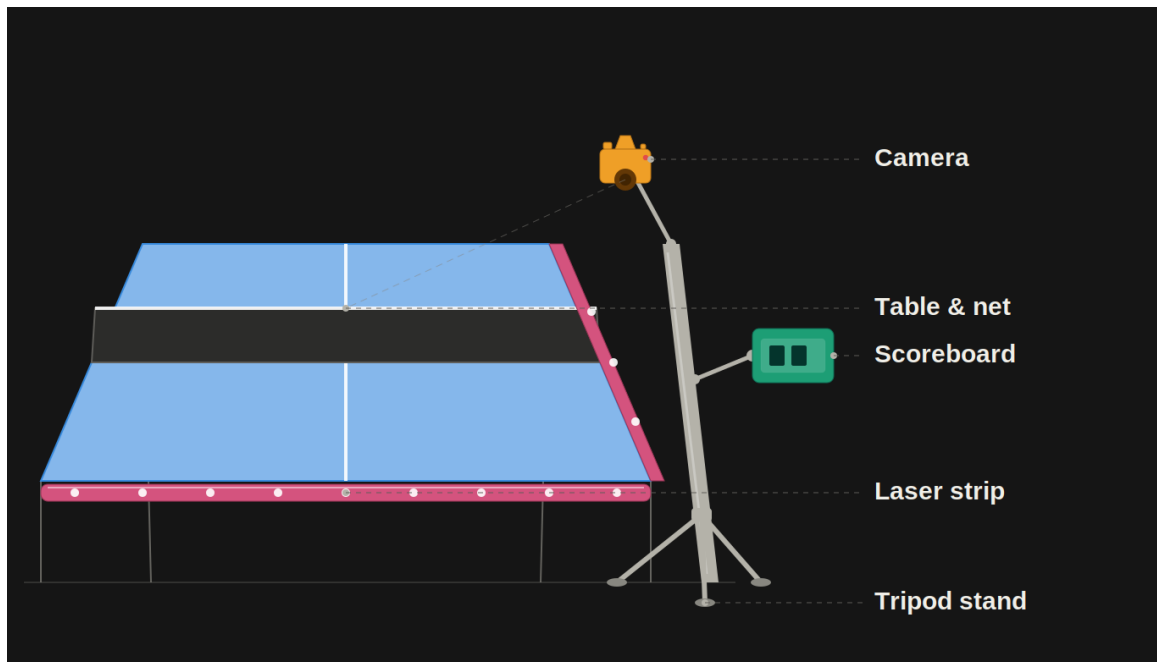


Figure 2.7.3.1: Prototype Illustration - Option 1 [6]

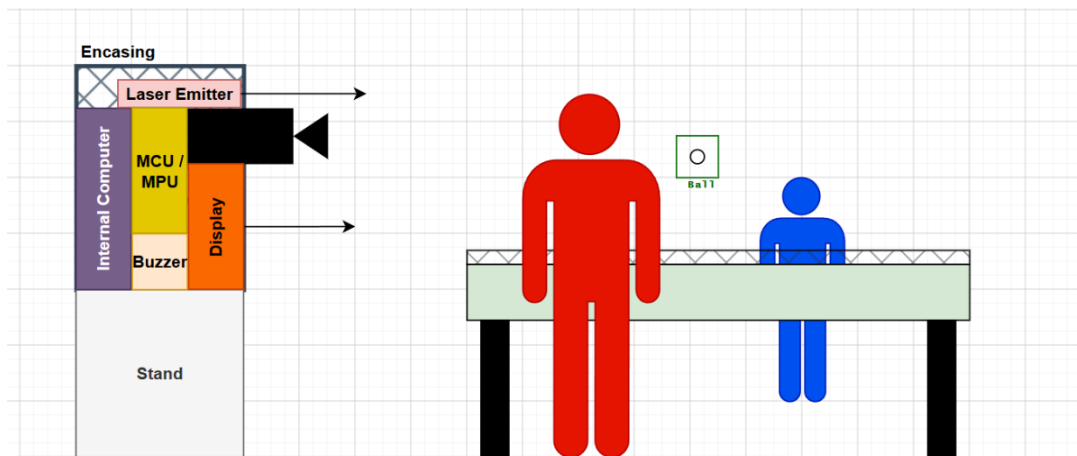


Figure 2.7.3.2: Prototype Illustration - Option 2

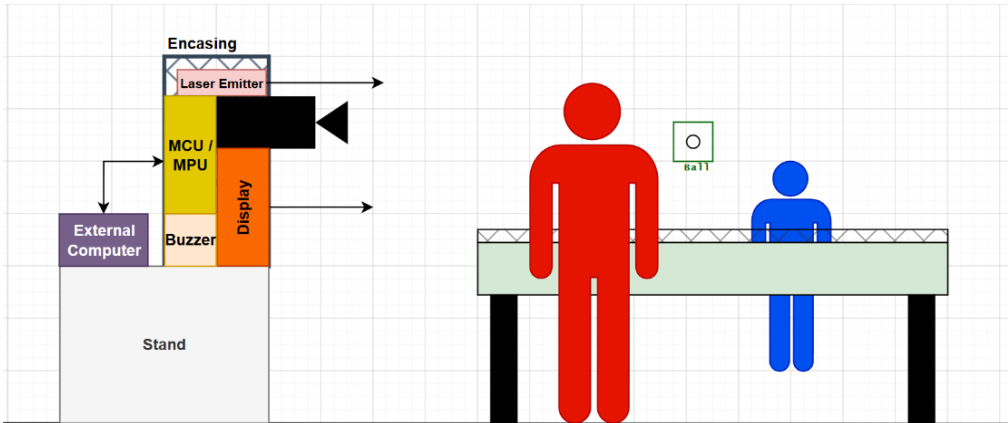


Figure 2.7.3.3: Prototype Illustration - Option 3

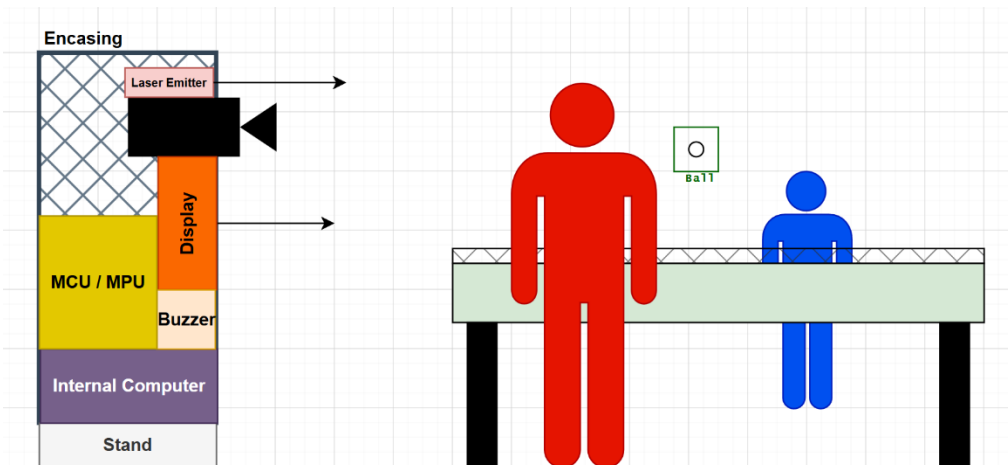


Figure 2.7.3.4: Prototype Illustration - Option 4

2.7.4 Laser Prototype / Blueprint

Figure 2.7.4 illustrates the laser grid to be included in the project. Each module represents 10 laser diodes and 10 photodiodes respectively.

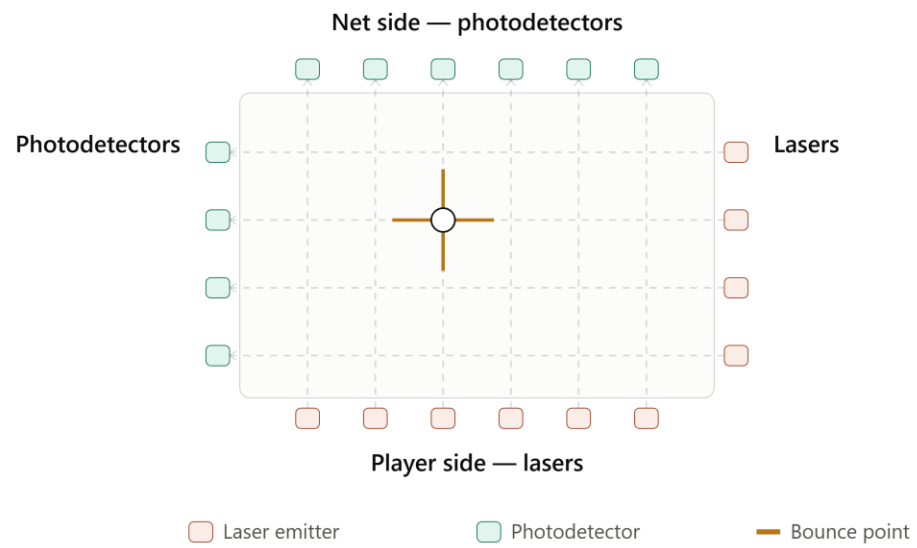


Figure 2.7.4: *Laser Grid Prototype Illustration*

2.8 House of Quality

The house of quality matrix serves as the visual representation of the correlation between the project's engineering and marketing requirements. As seen in Figure 2.8, the roof of the graphic also displays the correlation between each and every one of the engineering requirements.

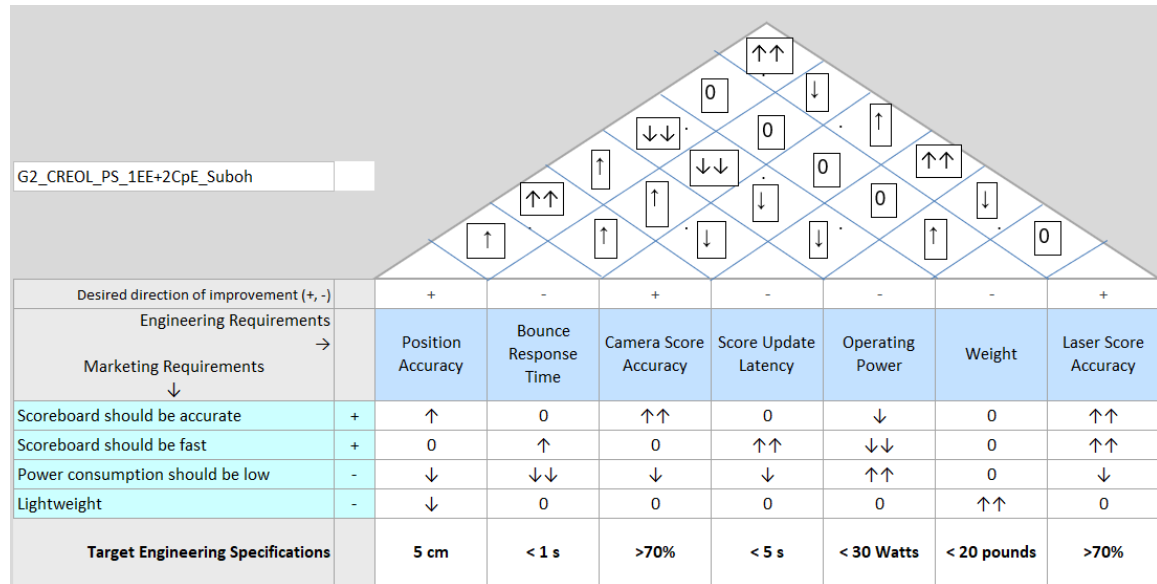


Figure 2.8: House of Quality

2.9 Technical Risks

The following table covers the three major technical risks that threaten the success of this project. Each risk has its own severity level, description, a list of effects caused by the failure of its implementation, and mitigation techniques the team hopes will minimize the chances of failure.

Table 2.9: Technical Risks

Risk Description	Risk Severity	Effect of Risk	Mitigation Technique
Laser array cannot track a ball's bounce fast enough.	Medium	The camera vision system will be the sole tracker of when the ball bounces. This is a significantly less accurate method of tracking when a bounce occurs. The scoreboard will no	Use photodiodes with a fast-enough frequency response to capture the change in light caused by the ball crossing the beam.

		longer be able to accurately award points, thus losing count of the score.	
Too many laser diodes and photodetectors in the system	Low	System is slowed down by the sheer number of laser diodes and photodetectors driving the laser array. This is paired with the system running the computer vision program, so the processing power would run thin.	Utilize a time-division multiplexer to cycle the signal of the lasers very fast to keep the number of channels used by the laser array low.
Computer vision software solution lacks accuracy	High	The ball tracking system is not accurate when the ball is moving through the air or when a player hits the ball. This can cause issues with determining who gets a point and can impact the advanced and stretch goals of showing the replay of the game.	Consult with committee members to ensure the software solution is accurate enough to use. Proper time usage and management will also allow more time to refine the software.

Chapter 11 - Administrative Content

11.1 Budget

The budget we set for this project consists of \$600. This project will be self-funded by the students. This amount was defined in accordance with the entire team and will be divided as follows:

Table 11.1: Project Budget

Sub-System	Budget	Description	Percentage
MPU	\$130	Main processing unit running the computer vision pipeline and game-state logic	21.6%
Power	\$70	Battery/AC-DC adapter, voltage regulators, and wiring to power all subsystems.	11.6%
Display	\$60	LED/LCD scoreboard panel and its driver electronics.	10%
Input Controls	\$10	Manual buttons/switches for score adjustment and reset.	1.6%
PCB(s)	\$50	Custom PCB fabrication for the sensor interface and driver board.	8.3%
Photonics	\$150	Laser diodes, photodetectors, and optics for the table bounce grid and net-line array.	25%
Camera	\$110	Camera module and lens for overhead ball tracking.	18.3%
Spotlight	\$10	Laser pointer module for highlighting the ball's last bounce location.	1.6%
Illumination	\$10	LED array and driver circuit for consistent table illumination.	1.6%
TOTAL	\$600	FULL PROJECT BUDGET	100%

11.2 Bill of Materials

[Not required for D&C submission]

11.3 Distribution of Work

The following section provides a high-level view of how each responsibility was distributed. The list is not comprehensive, and the project still relies on heavy collaboration to prevent blockers and other issues from affecting progress.

Table 11.3: Work Distribution

Category	Responsibility	Primary Member	Secondary Member
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Documentation	Demo Video + Presentation	Collaborative	
	Final Report	Collaborative	
	Website	Damian Orzejovsky	Ivan Wong
Design	PCB Design	Neel Patel	Ivan Wong
	PSU Design	Neel Patel	Ivan Wong
	Computer Vision	Damian Orzejovsky	Ivan Wong
	Low-Level Software	Ivan Wong	Damian Orzejovsky
	Laser Grid System	Patrick Suarez	Neel Patel
	Bounce Pointer	Patrick Suarez	Damian Orzejovsky

11.4 Milestones

In order to finalize the Final Report on time, the team designed the following milestone table to keep ourselves accountable and organized throughout the entire process. While the general dates may change, the goal is to stick to these as closely as possible while adjusting upcoming milestones based on previous outcomes.

11.4.1 Design Milestones

This table contains all design milestones to achieve all the way until the submission of the Final Report. Each entry has its own description, estimated beginning and completion dates, alongside other requirements.

Table 11.4.1: Project Design Milestones

Milestone Number	Milestone Description	From	To (Approx.)	Dependencies & Requirements
1	Finalize D&C Document	9/3/26	9/11/26	Review by committee
2	D&C Meeting	9/14/26		
3	Finalize D&C Revision	9/14/26	9/18/26	D&C submission feedback
<i>Start of 4 pages per week per member; 10 weeks to submission.</i>				
4	Define HW Specifications	9/14/26	9/21/26	Basic model testing and benchmarking.
5	30 Pages + Initial BOM	9/18/26	10/2/26	Should have ~42 pages
6	60 Pages	9/18/26	10/18/26	Should have ~74 pages
7	Midterm Milestone Report	11/6/26		Report should have ~106 pages

8	Midterm Report Revision	11/9/26	11/13/26	Midterm Report Feedback
9	120 Pages	9/18/26	11/18/26	Should have ~138
10	Mini Demo Video	11/1/26	12/3/26	Report should be near completion
11	Final Report	9/18/26	12/3/26	At least 120 pages

11.4.2 Fabrication Milestones

Similar to our design milestones (Section 11.4.1), the following table outlines key points in the fabrication process of the final prototype.

Table 11.4.2: Project Fabrication Milestones

Milestone Number	Milestone Description	From	To (Approx.)	Dependencies & Requirements
<i>Senior Design I</i>				
1	Initial Bill of Materials	9/18/26	10/2/26	Preliminary system specifications
2	Order Materials	10/2/26	10/9/26	BOM
3	Design Schematic	10/2/26	10/16/26	Must verify compatibility of all components
4	System Testing	10/16/26	11/16/26	All individual components must work
5	Photonic System Development	10/16/26	11/16/26	Photonic Laser grid must track bounces and position
<i>Senior Design II</i>				
6	PCB Design	1/11/27	2/11/27	Design initial PCB
7	PCB Testing	2/11/27	2/18/27	Test initial PCB and figure out improvements
8	PCB Refactoring + Polishing	2/18/27	3/4/27	Buffer time to fix issues with initial design and prototype polishing
9	Complete Prototype	3/4/27	4/1/27	Final prototype ready for demonstration

Appendices

Appendix A – References

- [1] B. Church, "CNN," 9 July 2025. [Online]. Available: <https://www.cnn.com/2025/07/09/sport/wimbledon-ball-boy-electronic-line-call-spt-intl>. [Accessed 10 September 2026].
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- [3] R. Jain, "TableTennisTracker," 2022. [Online]. Available: <https://github.com/jainr3/TableTennisTracker>. [Accessed 7 September 2026].
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- [6] Anthropic, *Claude Sonnet 5 (large language model)*, 2026.

Appendix B – Copyright Permission

[N/A]